

## AP CALCULUS AB – UNIT 2

## Differentiation: Definition and Fundamental Properties

AP-Style Review and Exam Practice



Name: \_\_\_\_\_ Class: \_\_\_\_\_ Date: \_\_\_\_\_

## Directions

Section I Part A contains 16 non-calculator multiple-choice questions. Section I Part B contains 8 calculator-permitted multiple-choice questions. Section II contains four free-response questions. Show setup, exact reasoning, units, and justification.

## Section I – Part A: No Calculator

1. For  $f(x) = x^2 - 4x$ , find the average rate of change on  $[4, 5]$ .

- (A)  $-5$  (B)  $6$   
(C)  $5$  (D)  $4$

2. If  $f(x) = 2x^2 - 1x + 1$ , evaluate  $\lim_{h \rightarrow 0} \frac{f(3+h) - f(3)}{h}$ .

- (A)  $11$  (B)  $-11$   
(C)  $10$  (D)  $12$

3. Which limit represents  $f'(2)$ ?

- (A)  $\lim_{h \rightarrow 2} \frac{f(h) - f(2)}{h}$  (B)  $\lim_{h \rightarrow 0} \frac{f(2) - f(h)}{2 - h}$   
(C)  $\lim_{h \rightarrow 0} \frac{f(2+h) - f(2)}{h}$  (D)  $\lim_{x \rightarrow 2} f(x)$

4. Find the tangent line to  $y = x^2 - 2x + 3$  at  $x = -2$ .

- (A)  $y = 23 - 6x$  (B)  $y = 1 - 5x$   
(C)  $y = -6x$  (D)  $y = -6x - 1$

5. A table gives  $f(\frac{39}{10}) = -\frac{68}{5}$  and  $f(\frac{41}{10}) = -\frac{72}{5}$ . Estimate  $f'(4)$  using a symmetric difference.

- (A)  $-\frac{4}{5}$  (B)  $-4$   
(C)  $-2.0$  (D)  $-8$

6. Which condition is sufficient to conclude that  $f$  is continuous at  $x = c$ ?

- (A)  $f(c)$  exists (B)  $\lim_{x \rightarrow c} f(x)$  exists  
(C)  $f$  has equal one-sided derivatives at  $c$  but is undefined there (D)  $f$  is differentiable at  $x = c$

7. Differentiate  $f(x) = 3x^5 + 2x^2 - 3$ .

- (A)  $x(15x^4 + 4)$  (B)  $15x^4 + 4x + 1$   
(C)  $x(15x^3 + 4)$  (D)  $x(3x^3 + 4)$

8. Find  $\frac{d}{dx}[\log(x)]$ .

(A)  $\frac{1}{x}$

(B)  $-\frac{1}{x}$

(C)  $\log(x) + 1$

(D)  $x$

9. Differentiate  $(x^2 + 1)(2x - 1)$ .

(A)  $4x$

(B)  $2x(2x - 1)$

(C)  $6x^2 - 2x + 2$

(D)  $2x^2 + 2$

10. Differentiate  $\frac{x^2 + 1}{x + 2}$ .

(A)  $\frac{-x^2 + 2x(x+2) - 1}{(x+2)^2}$

(B)  $\frac{x^2 + 2x(x+2) + 1}{(x+2)^2}$

(C)  $2x$

(D)  $\frac{2x}{x+2}$

11. At  $x = 2$ ,  $f = 4$ ,  $g = 3$ ,  $f' = 1$ , and  $g' = 3$ . Find  $(fg)'(2)$ .

(A) 14

(B) -15

(C) 16

(D) 15

12. At  $x = 1$ ,  $f = 4$ ,  $g = 3$ ,  $f' = 1$ , and  $g' = -3$ . Find  $(f/g)'(1)$ .

(A)  $-\frac{5}{3}$

(B)  $\frac{2}{3}$

(C)  $\frac{8}{3}$

(D)  $\frac{5}{3}$

13. For  $f(x) = 3x^2 + x$ , find the average rate of change on  $[1, 2]$ .

(A) 11

(B) 10

(C) 9

(D) -10

14. If  $f(x) = 3x^2 + 3x + 1$ , evaluate  $\lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h}$ .

(A) 3

(B) 4

(C) 2

(D) -3

15. Which limit represents  $f'(1)$ ?

(A)  $\lim_{x \rightarrow 1} f(x)$

(B)  $\lim_{h \rightarrow 1} \frac{f(h) - f(1)}{h}$

(C)  $\lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h}$

(D)  $\lim_{h \rightarrow 0} \frac{f(1) - f(h)}{1-h}$

16. Find the tangent line to  $y = x^2 - x + 1$  at  $x = 1$ .

(A)  $y = x + 1$

(B)  $y = 2x - 1$

(C)  $y = x$

(D)  $y = x + 2$

**Section I – Part B: Calculator Permitted**

- 17.** Use a calculator to evaluate the instantaneous rate of change of  $f(x) = 1.35x^2 - 0.17x$  at  $x = 2.59$ .  
(A) 7.823 (B) 6.823  
(C) 8.616 (D) 2.53
- 18.** Use a calculator to find  $f'(0.38)$  for  $f(x) = \sin x + e^x$ .  
(A) 1.484 (B) 1.391  
(C) 2.391 (D) 1.833
- 19.** A tangent to  $y = \ln x$  is drawn at  $x = 2.73$ . What is its slope to three decimal places?  
(A) 0.366 (B) 2.73  
(C) 1.004 (D) -0.366
- 20.** A calculator table gives  $f(2.4) = 1.4$ ,  $g(2.4) = 3.5$ ,  $f'(2.4) = -2.2$ , and  $g'(2.4) = 0.7$ . Find  $(fg)'(2.4)$ .  
(A) 6.72 (B) -6.72  
(C) -7.72 (D) -5.72
- 21.** A population is  $P(t) = 3.9e^t$  thousand. What is the instantaneous growth rate at  $t = 0.87$ ?  
(A) 9.309 thousand (B) 9.309 thousand per unit time  
(C) 8.099 thousand per unit time (D) 3.393 thousand per unit time
- 22.** Use a calculator to evaluate the instantaneous rate of change of  $f(x) = 1.54x^2 - 0.66x$  at  $x = 2.46$ .  
(A) 7.917 (B) 6.917  
(C) 2.42 (D) 7.696
- 23.** Use a calculator to find  $f'(1.88)$  for  $f(x) = \sin x + e^x$ .  
(A) 12.016 (B) 7.506  
(C) 6.249 (D) 5.249
- 24.** A tangent to  $y = \ln x$  is drawn at  $x = 1.5$ . What is its slope to three decimal places?  
(A) 1.5 (B) 0.667  
(C) 0.405 (D) -0.667

## Section II – Free Response

**Questions 1–2: Calculator permitted. Questions 3–4: No calculator.**

### FRQ1 – Rate of change and tangent line

Let  $f(x) = x^3 - 4x + 2$ .

(a) Write  $f'(2)$  as a limit of a difference quotient. [2 points]

(b) Find  $f'(2)$  exactly. [2 points]

(c) Write the tangent-line equation at  $x = 2$ . [2 points]

### FRQ2 – Derivative values from a table

Differentiable functions  $f$  and  $g$  have  $f(1) = 3$ ,  $g(1) = 2$ ,  $f'(1) = -1$ , and  $g'(1) = 4$ .

(a) Find  $(fg)'(1)$ . [2 points]

(b) Find  $(f/g)'(1)$ . [3 points]

(c) Interpret  $f'(1) = -1$  if  $f$  is measured in liters and  $x$  in minutes. [2 points]

**FRQ3 – Differentiability and continuity**

A function is defined by  $p(x) = ax + 3$  for  $x < 2$  and  $p(x) = x^2 + b$  for  $x \geq 2$ .

(a) Find a relation between  $a$  and  $b$  that makes  $p$  continuous at 2. [2 points]

(b) Find  $a$  and  $b$  that make  $p$  differentiable at 2. [3 points]

(c) Explain why differentiability automatically settles continuity. [1 points]

**FRQ4 – Mixed derivative rules**

$$\text{Let } q(x) = \frac{(x^2 + 1)e^x}{x + 2}.$$

(a) Find  $q'(x)$  without expanding. [4 points]

(b) Evaluate  $q'(0)$ . [2 points]

(c) Write the tangent line at  $x = 0$ . [2 points]